

Software Architecture and Design Specification

Project title: Online Internship & Placement Preparation Portal

Version: 1.0

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1.0	03-10-2026	Patnaikuni Sai Dheeraj, Sohel Mahibubasa Chapparband, Navya R, Nikil Raj G	Initial Software Architecture and Design Specification

Approvals

Role	Name	Signature/Date
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1. Introduction

1.1 Purpose

This document specifies the architecture and design of the **Online Internship & Placement Preparation Portal**.

It describes the major software components, architecture pattern, security architecture, requirements traceability, UML sequence diagrams, API design, and error-handling mechanisms of the system.

1.2 Scope

The system provides an online platform for students to:

- Register and log in.
- Create and manage their profiles.
- Search and filter internship opportunities.
- Apply for available internships.
- Track internship application status.
- Take aptitude tests.
- View test scores and test history.
- Record preparation progress.

- View preparation analytics.

The system also provides functionality for:

- **Recruiters** to create and manage internship listings and applications.
- **Administrators** to manage users, internships, aptitude tests, and questions.

1.3 Audience

This document is intended for:

- Developers
- QA engineers
- Software Engineering faculty
- Project team members
- Testers
- Maintenance teams

1.4 Definitions

Term	Definition
MERN	MongoDB, Express.js, React.js, and Node.js
REST	Representational State Transfer
API	Application Programming Interface
RBAC	Role-Based Access Control
UI	User Interface
SRS	Software Requirements Specification
STP	Software Test Plan
RTM	Requirements Traceability Matrix

2.Document Overview

2.1 How to Use This Document

This document provides the architectural and detailed design specification of the Online Internship & Placement Preparation Portal.

It includes:

- Architecture overview
- UML component diagram
- Component descriptions
- Architecture pattern
- Requirements traceability
- Security architecture
- UML sequence diagrams
- API design
- Error handling
- Database design considerations

2.2 Related Documents

The following project documents are related to this specification:

1. Software Requirements Specification (SRS)
2. Software Test Plan (STP)
3. Requirements Traceability Matrix (RTM)

3.Architecture

3.1 Goals & Constraints

Goals

The architecture aims to provide:

- Secure user authentication and authorization.
- Clear separation between frontend, backend, and database.
- Support for Student, Recruiter, and Admin roles.
- Maintainable and modular software components.
- Reliable internship application processing.
- Aptitude test evaluation and result storage.
- Preparation progress and analytics.

Constraints

- The system requires network connectivity.
- The frontend is implemented using React.js.
- The backend uses Node.js and Express.js.
- MongoDB is used for persistent data storage.
- Protected operations require authentication.
- Role-specific operations require appropriate authorization.

3.2 Stakeholders & Concerns

Stakeholder	Main Concerns
Students	Easy internship search, applications, tests, progress tracking
Recruiters	Internship management and application management
Administrators	User, internship, test, and question management
Developers	Modularity, maintainability, and clear component separation
Testers	Correct functionality, security, performance, and usability
Faculty	Requirement coverage and appropriate architecture/design

3.3 Component (UML) Diagram

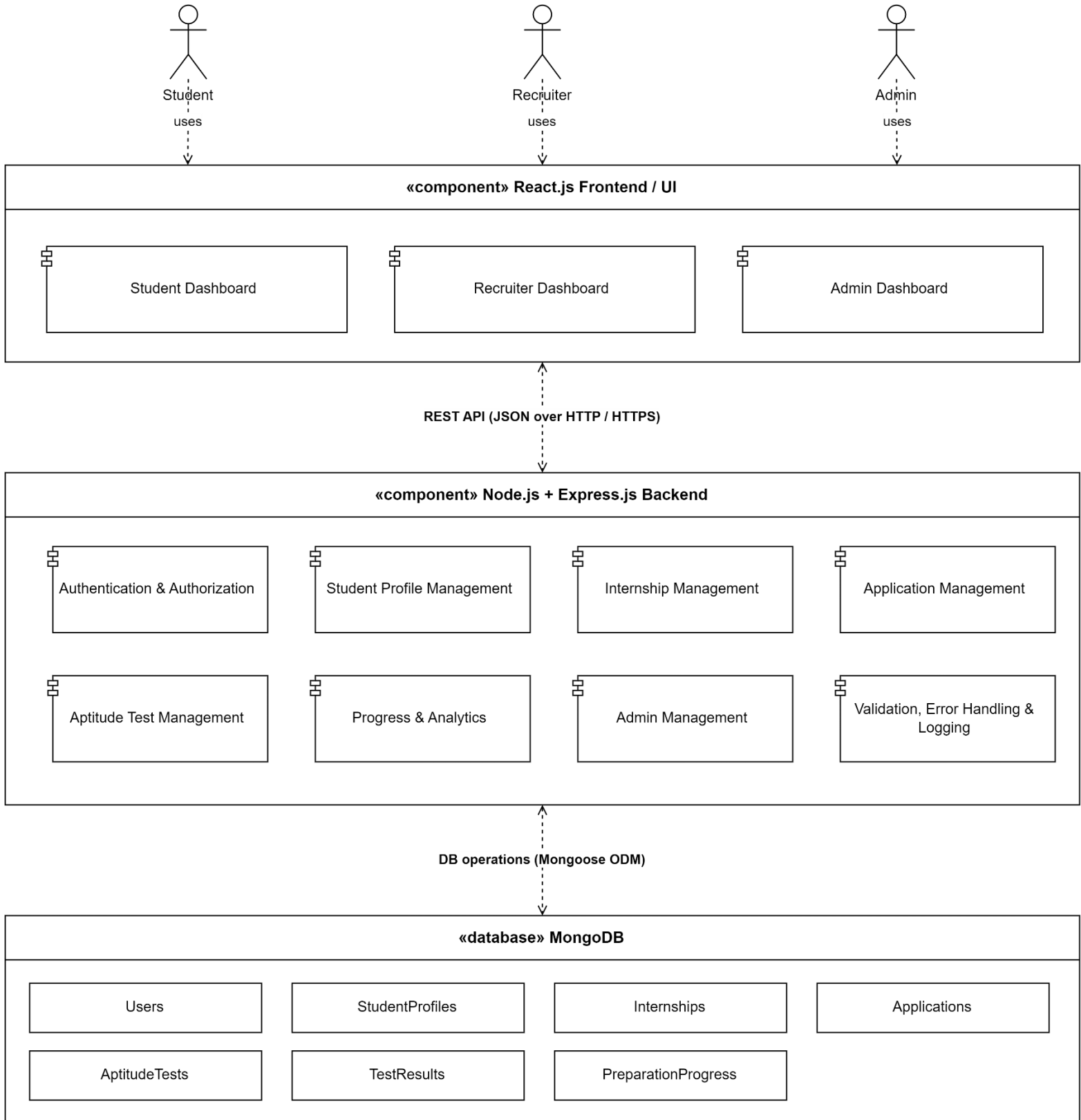


Fig. 1. UML Component Diagram of the Online Internship & Placement Preparation Portal

3.4 Component Descriptions

Component	Description
React.js Frontend / UI	Provides the user interface for students, recruiters, and administrators.
Authentication & Authorization	Handles user authentication and role-based access control.
Student Profile Management	Handles student personal, academic, skill, and resume-related information.
Internship Management	Displays, searches, filters, creates, updates, and manages internship listings.
Application Management	Handles internship applications, duplicate-application checks, and application status.
Aptitude Test Management	Provides available tests, accepts answers, calculates scores, and stores results.
Progress & Analytics	Records preparation progress and provides progress and test-result analytics.
Admin Management	Allows administrators to manage users, internships, tests, and questions.
MongoDB	Stores user, profile, internship, application, test, result, and progress information.

3.6 Technology Stack & Data Stores

Category	Technology
Frontend	React.js
Backend	Node.js
Backend Framework	Express.js
Database	MongoDB
API Style	REST API
Data Format	JSON
Communication	HTTP/HTTPS

3.7 Risks & Mitigations

Risk	Mitigation
Backend or database unavailable	Handle failures and display appropriate error messages.
Unauthorized access	Authentication and role-based authorization.
Invalid user input	Frontend and backend validation.
Duplicate internship application	Check existing application before creating a new one.
High number of concurrent users	Performance and load testing according to SRS requirements.

3.8 Traceability to Requirements

Requirement	Component
IPR-F-001 – Student registration	Authentication & Authorization
IPR-F-002 – Authentication	Authentication & Authorization
IPR-F-003 – Role-based access	Authentication & Authorization
IPR-F-004 – Student profile	Student Profile Management
IPR-F-006 – Display internships	Internship Management
IPR-F-007 – Search/filter internships	Internship Management
IPR-F-008 – Apply for internship	Application Management
IPR-F-009 – Prevent duplicate applications	Application Management
IPR-F-010 – Application status	Application Management
IPR-F-011 – Aptitude tests	Aptitude Test Management
IPR-F-012 – Calculate score	Aptitude Test Management
IPR-F-013 – Test history	Aptitude Test Management
IPR-F-014 – Preparation progress	Progress & Analytics
IPR-F-015 – Analytics	Progress & Analytics
IPR-F-016 – Recruiter internship management	Internship Management
IPR-F-017 – Recruiter application management	Application Management
IPR-F-018 – Admin management	Admin Management

3.9 Security Architecture

The system includes the following security mechanisms:

Authentication

Users must authenticate before accessing protected functionality.

Password Protection

Passwords should be stored using secure password hashing rather than plaintext storage.

Role-Based Access Control

The system supports three roles:

- Student
- Recruiter
- Admin

Each role is permitted to access only the functionality associated with that role.

HTTPS/TLS

HTTPS/TLS should be used to protect communication between the frontend and backend in deployment.

Input Validation and Sanitization

User input should be validated before processing or storing it in the database

4. Design

4.1 Design Overview

The system is designed using modular frontend and backend components.

The frontend communicates with backend services through REST APIs. The backend performs authentication, authorization, business processing, validation, and database operations.

4.2 UML Sequence Diagrams

Sequence Diagram 1 — Student Login

Student → React UI → Express Backend → Authentication → MongoDB

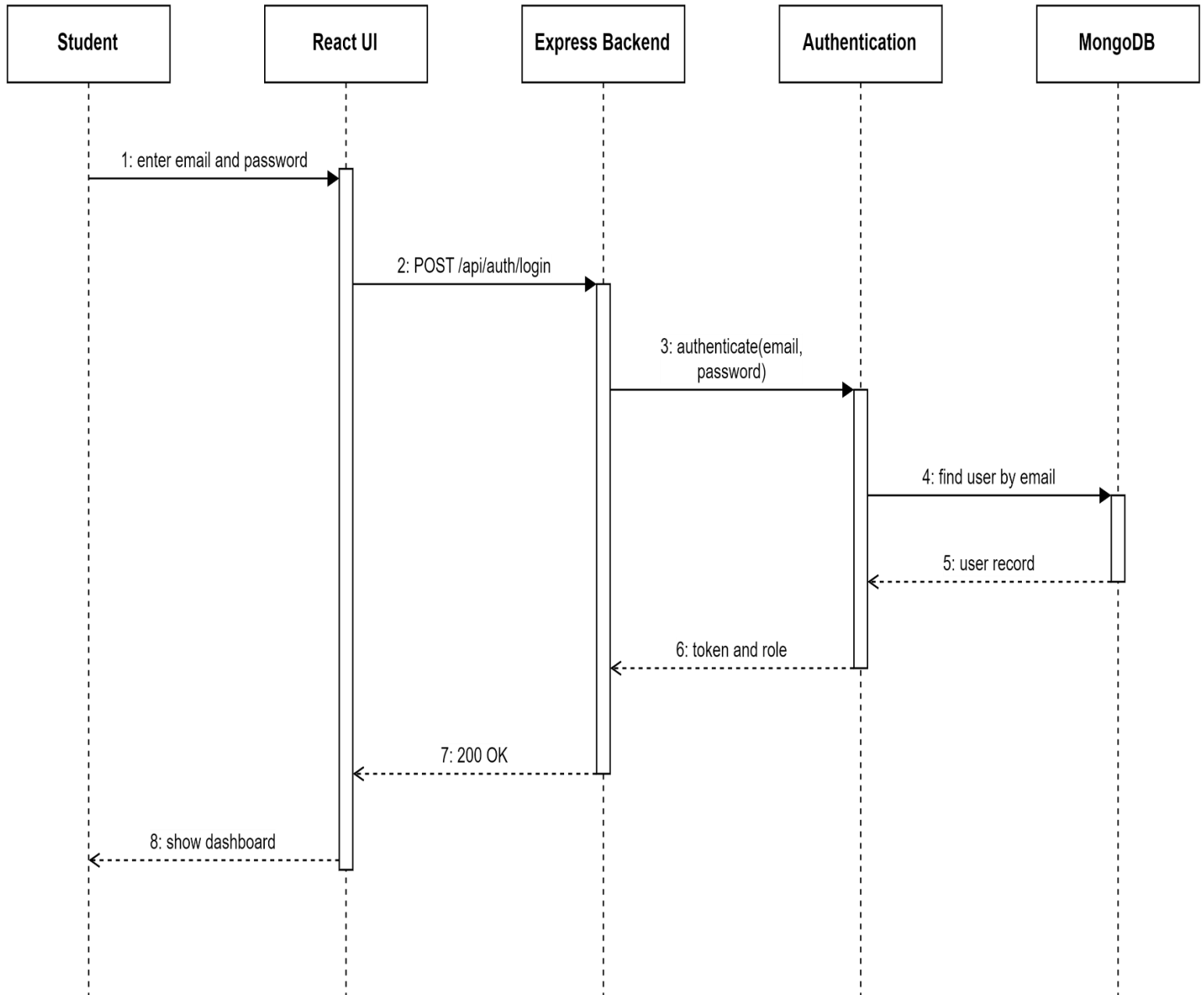


Fig. 2. UML Sequence Diagram - Student Login

Sequence Diagram 2 — Student Internship Application

Student → React UI → Express Backend → Application Management → MongoDB

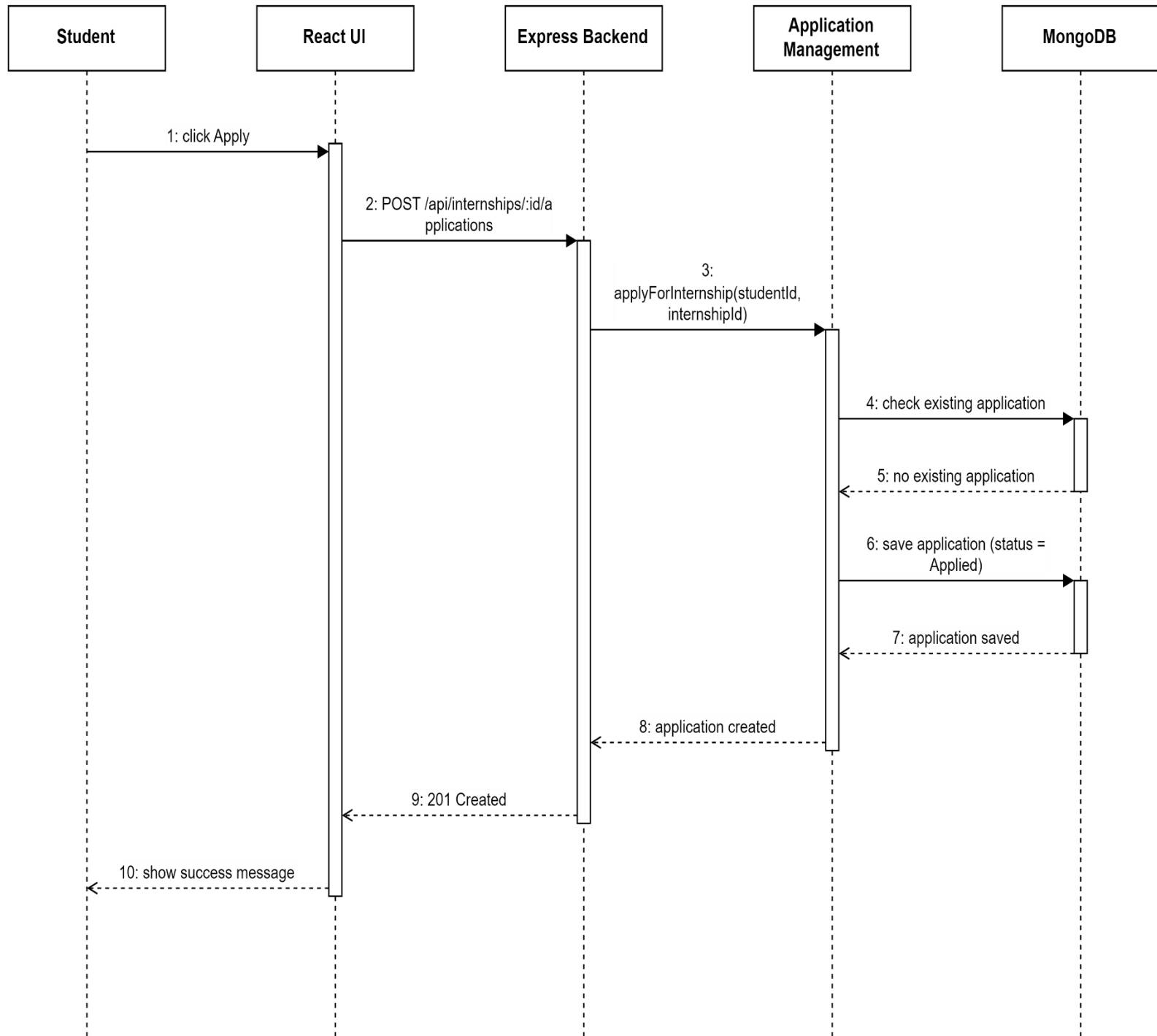


Fig. 3. UML Sequence Diagram - Student Internship Application

4.3 API Design

Method	Endpoint	Purpose
POST	<code>/api/auth/register</code>	Register a new user
POST	<code>/api/auth/login</code>	Authenticate user and provide access
GET	<code>/api/internships</code>	View available internships
GET	<code>/api/internships/:id</code>	View internship details
POST	<code>/api/internships/:id/applications</code>	Apply for an internship
GET	<code>/api/applications/me</code>	View student's application status
GET	<code>/api/tests</code>	View available aptitude tests
POST	<code>/api/tests/:id/submit</code>	Submit an aptitude test
GET	<code>/api/test-results/me</code>	View student's test results and history
GET	<code>/api/progress/me</code>	View preparation progress
PUT	<code>/api/progress/me</code>	Update preparation progress

4.4 Error Handling, Logging & Monitoring

Status	Meaning
400	Invalid request/input
401	Authentication failure
403	Insufficient permissions
404	Resource not found
409	Conflict, such as duplicate application
500	Internal server error

4.5 UX Design

The portal provides separate interfaces according to the user's role.

Student

- Internship search
- Internship application
- Application tracking
- Aptitude tests
- Test results
- Preparation progress
- Analytics

Recruiter

- Internship creation
- Internship management

- Application viewing
- Application status updates

Admin

- User management
- Internship management
- Aptitude test management
- Question management

4.6 Open Issues & Next Steps

- **API Endpoints:** Finalize the REST API endpoints based on the implemented backend.
- **Authentication:** Confirm the authentication and authorization mechanism.
- **Database:** Finalize the MongoDB collections and required fields.
- **UML Diagrams:** Create and verify the component and sequence diagrams.
- **Testing:** Perform integration and system testing to verify system functionality.
- **Security:** Verify authentication, authorization, input validation, and other security requirements.
- **Performance:** Conduct performance testing to validate response time and concurrent-user requirements.

4. Appendices

5.1 Glossary:

Term	Meaning
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RBAC	Role-Based Access Control
SRS	Software Requirements Specification
STP	Software Test Plan
RTM	Requirements Traceability Matrix
UI	User Interface

5.2 References:

- **ISO/IEC/IEEE 42010:2022**, *Software, systems and enterprise — Architecture description*, IEEE/ISO/IEC, 2022.
- [2] **MongoDB Documentation**, *Integrate MongoDB with React*, MongoDB.
- [3] **OWASP Foundation**, *REST Security Cheat Sheet*, OWASP.
- [4] **OWASP Foundation**, *Authentication Cheat Sheet*, OWASP.

5.3 Tools:

- **Draw.io** – Used to create UML and architecture diagrams.
- **Visual Studio Code** – Used for coding and project development.
- **MongoDB Tools** – Used to manage and inspect the MongoDB database.